

2. ROS2 Installation (Jazzy)

Note: The accompanying reference video uses the ROS Humble version, for reference only

- ROS2 Jazzy corresponds to Ubuntu 24.04
- To install other ROS2 versions, simply replace **jazzy** in the installation commands with the desired version. This applies to all ROS2 version-specific software in the ROS2 tutorial series. Subsequent tutorials will not repeat this explanation.
- **Yahboom image already has the basic ROS environment installed. Related packages may need to be downloaded separately**

1. Setting Up Locale

Please check if your locale supports UTF-8 encoding first. Use the following commands to check and set UTF-8 encoding:

```
locale # Check UTF-8 support

sudo apt update && sudo apt install locales
sudo locale-gen en_US en_US.UTF-8
sudo update-locale LC_ALL=en_US.UTF-8 LANG=en_US.UTF-8
export LANG=en_US.UTF-8

locale # Verify settings
```

Note: The locale can be different, but it must support UTF-8 encoding.

2. Installing Jazzy

First, update the apt repository cache:

```
sudo apt update
```

Then upgrade installed software (ROS2 packages are built on frequently updated Ubuntu systems. Please ensure your system is up to date before installing new packages):

```
sudo apt upgrade
```

Install ROS2 Jazzy Desktop (recommended), which includes: ROS, RViz, examples and tutorials. The installation command is as follows. If you need to install other versions, replace "jazzy" with the corresponding version number in the command below:

```
sudo apt install ros-jazzy-desktop python3-argcomplete
```

Install colcon build tools:

```
sudo apt install python3-colcon-common-extensions
```

3. Configuring Environment

- For other ROS2 versions, replace jazzy with the corresponding version identifier. For example, for Humble, replace jazzy with humble.

When running ROS2 programs in the terminal, you need to source the environment:

```
source /opt/ros/jazzy/setup.bash
```

You need to execute the above command every time you open a new terminal, or you can run the following command to write the environment setup to `~/.bashrc`, so you don't need to manually configure the environment each time you start a new terminal:

```
echo "source /opt/ros/jazzy/setup.bash" >> ~/.bashrc
```

At this point, ROS2 has been installed and configured.

4. Running Publisher and Subscriber Example Nodes

ROS2 desktop comes with pre-installed example nodes. We can test if ROS2 is installed successfully by running these example nodes.

- Run the publisher node:

```
ros2 run demo_nodes_cpp talker
```

- Run the subscriber node:

```
ros2 run demo_nodes_cpp listener
```

5. Running the TurtleSim Demo

Turtlesim is a classic teaching tool in ROS. Through turtlesim, you can quickly understand the core concepts in ROS.

- Run turtlesim:

```
ros2 run turtlesim turtlesim_node
```

- Run keyboard control node:

```
ros2 run turtlesim turtle_teleop_key
```

- View velocity topic data:

```
ros2 topic echo /turtle1/cmd_vel
```

6. Uninstalling

After ROS2 is installed, if you want to uninstall ROS2, run the following commands:

```
sudo apt remove ~nros-jazzy-* && sudo apt autoremove
```

You can also remove the ROS2 repository:

```
sudo rm /etc/apt/sources.list.d/ros2.list  
sudo apt update  
sudo apt autoremove
```